German University in Cairo

Faculty of Information Engineering and Technology
Dr. Mohammad Shehab

MATH502
Assignment 1

- Each group consists of 2 students (cross tutorials is allowed).
- There will be an in-person evaluation for each member.
- Deadline for submission: 6th of November. Each delay of 1 day will be subject to 1 mark deduction

Deliverables are expected to be as follows:

1. A cover page listing the students name and ID numbers.
2. The use of a single programming language of choice (Wolfram Mathematica (recommended), Mathworks Matlab, Python, etc…), with appropriate justification.
3. Provide all analysis, used code, and the corresponding output results, with appropriate textual commenting.
4. You can send a google drive link that includes your submission package (folder) to:
   bahaaeldin.ibrahim@guc.edu.eg
   All submitted files must be included in the link

Problem 1

The LORAN (LOng RAnge Navigation) system calculates the position of a boat at sea using signals from fixed transmitters. From the time differences of the incoming signals, the boat obtains differences of distances to the transmitters. This leads to two equations each representing hyperbolas defined by the differences of distance of two points (foci). An example of such equations are:

$$\begin{aligned} \frac{x^2}{186^2} - \frac{y^2}{300^2 - 186^2} &= 1, \\ \frac{(y-500)^2}{279^2} - \frac{(x-300)^2}{500^2 - 279^2} &= 1. \end{aligned} \tag{1}$$

Solving two quadratic equations with two unknowns would require solving a 4th degree polynomial equation. We could do this by hand, but for a navigational system to work well, it must do the calculations automatically and numerically. We note that the Global Positioning System (GPS) works on similar principles and must do similar computations.

Solve the non-linear system in (1) using any version you like of the Newton Raphson method. A good initial guess could be $x_0 = y_0 = 200$.

a) Provide at least two iterations of the solution using a pen and paper, showing all steps.

b) Use a software package to find accurate solution(s).

Problem 2

In a smart building, a Wireless Sensor Network (WSN) is deployed for environmental monitoring. To determine the signal strength (RSSI – Received Signal Strength Indicator) received by a node at a particular location, communication engineers collect measurements at various distances from a transmitter.

Due to environmental factors like wall interference and multipath reflection, the signal strength does not follow a simple model. Instead, the collected data must be used to interpolate the signal strength at locations where no measurements were taken.

You are given the following data of distance [m] from the transmitter and the corresponding RSSI [dBm]:

Distance [m]	RSSI [dBm]
1.	−40
2.5	−47
4.	−54
6.	−60
8.	−66

a) Using Newton's divided difference method, construct the interpolating polynomial that models the RSSI as a function of distance.

b) Use this polynomial to estimate the RSSI at 5 meters.

c) Compare this interpolated value with a measured value of -57 dBm at 5 meters (obtained during a different measurement session). Discuss possible reasons for any discrepancy.

d) Represent the polynomial using Horner's scheme, and state the number of additions and the number of multiplications carried out in this form.

Problem 3

Consider a simple DC circuit where a silicon diode is in series with a resistor and a DC voltage source. The equation governing the circuit is derived from Kirchhoff's Voltage Law (KVL):

$$V_s = I\,R + V_D \tag{2}$$

But the diode voltage V_D and current I are related nonlinearly by the Shockley diode equation:

$$I = I_s\left(e^{V_D/(n\,V_T)} - 1\right) \tag{3}$$

where

- $I_s = 1 \times 10^{-12}$ A (reverse saturation current)
- $n = 1.5$ (ideality factor)

- $V_T = 25.85 \times 10^{-3}$ V (thermal voltage at room temperature)
- $R = 1$ kΩ
- $V_s = 1.0$ V

The goal is to find the current I flowing through the circuit, which satisfies the nonlinear equation:

$$V_s = I\,R + n\,V_T \ln\left(\frac{I}{I_s} + 1\right) \tag{4}$$

Equation (5) cannot be solved algebraically. You must use a root-finding method to solve:

$$f(I) = I\,R + n\,V_T \ln\left(\frac{I}{I_s} + 1\right) - V_s = 0 \tag{5}$$

a) Define the function $f(I)$ as above using the given constants.

b) Use each of the root-finding methods (Bisection, and Secant) to find the root of $f(I)$, i.e., the value of I for which $f(I) = 0$, accurate to within 10^{-4} A. Use an initial bracket such as: a lower bound is $I = 0$, and an upper bound $I = 5 \times 10^{-3}$ A.

c) Report the number of iterations needed for each method, and final value of the current I.

d) Plot $f(I)$ over the interval and show where the root lies. Your plot should be neat, with clearly labelled axes.

Problem 4

Repeat Problem 4, using Newton's iterative method for root finding. Report the needed number of iterations for each of the following initial guesses for the current I:

a) $I = 0.35$ mA

b) $I = 1.0$ mA

Problem 5

A networking engineer is analyzing the latency of a packet traveling through a network as a function of packet size x [kB]. Due to time constraints, the engineer collects latency data [ms] at only a few discrete packet sizes. The goal is to model the latency function using polynomial interpolation, so that latency can be predicted for other packet sizes not directly measured.

The engineer gathers the following data:

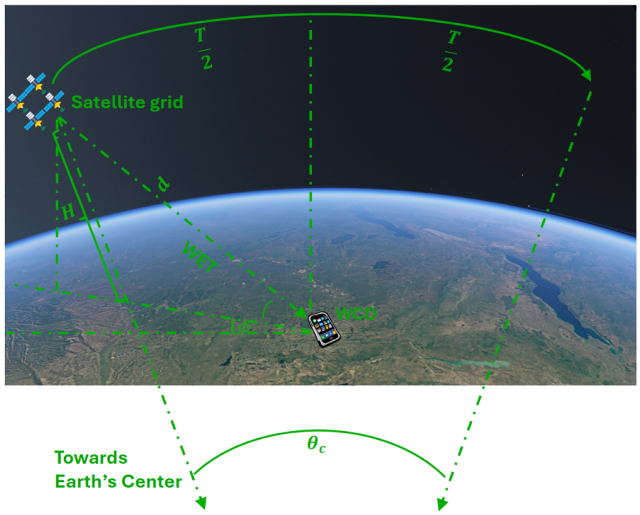
Packet size [kB]	Latency [ms]
150.	2.1
200.	2.85
230.	3.35
260.	3.95
300.	4.9
340.	6.1
400.	8.05

a) Find a polynomial function $L(x)$ that interpolates the latency values as a function of packet size x. Use the Vandermonde matrix approach to determine the coefficients of the interpolating polynomial.

b) Use this model to estimate the latency for a 3 KB packet.

Problem 6: Satellite to ground energy harvesting

Consider a Wirelessly Chargeable device (WCD) with a charging duration, which spans the time of rotation of the closest satellite grid from the start until the end of the charging visibility range of the WCD during the energy harvesting process. During this duration, the device can observe the satellite grid, which constitutes N constellation satellites moving together as shown in the figure below. Each satellite is equipped with M antennas.



The distance between the satellite grid and the WCD is given by

$$d(t) = \sqrt{(R+H)^2 + R^2 - 2(R+H)R\cos\phi\cos(\omega t)}.$$

The received power at the WCD is

$$P_r(t) = P_t G_T G_R \frac{\lambda^2}{(4\pi d(t))^2} \alpha_s \beta_s.$$

where the satellite channel Rician fading parameters are defined as:

$$\alpha_s = \frac{m(2b_0 + \Omega)^2}{4mb_0^2 + 4mb_0\Omega + \Omega^2},$$

$$\beta_s = \frac{4mb_0^2 + 4mb_0\Omega + \Omega^2}{m(2b_0 + \Omega)},$$

P_t is the satellite transmit power, λ is the wavelength. Then the harvested power is

$$P_h(t) = \begin{cases} 0, & P_r(t) < P_{th}, \\ \eta_h P_r(t), & P_r(t) \geq P_{th}, \end{cases}$$

$$T = \theta_c / \omega,$$

θ_c is the cut-off angle after which P_r is less than P_{th}

Due to the presence of satellites and antennas per satellite, the amount of harvested energy can be multiplied by MN^2, when applying maximum ratio combining. Thus, the total amount of harvested energy at the earth device is theoretically upper bounded by

$$E_h = MN^2 \int_{t=0}^{T} P_h(t)\, dt$$

1. Is there a closed form solution for this integral? If yes, attempt to obtain it

2. Write a script that evaluates this integral and the true error using:

a. Multiple Trapezoidal rule
b. One of the Simpson's rules
c. Monte-Carlo

3. Plot the harvested energy at the WCD for different number of satellites (4 to 32 with a step of 4) using numerical methods in a,b,c vs the true integral evaluation

System parameters:
$\phi = 0, M = 8, P_{th} = -20$ dBm

Parameter	Value	Parameter	Value
R	6378 Km	H	200 Km
G	$6.67\times 10^{-11}\frac{m^3}{Kg\ s^2}$	M_e	5.97×10^{24} Kg
G_T	50 dB	G_r	10 dB
Ω	1.29	f	868 MHz
m	19.4	P_t	40 dBm
b_o	0.158	η_h	70%

ω: satellite angular velocity
H: Satellite grid height
M_e: Mass of earth
G: gravitational constant
R: Earth's radius
G_T, G_r: Transmit and receive antenna gain
η_h is the WCD energy harvesting efficiency